Trip-Length Distributions Enable Recovery of Distance-Decay Parameters for Zero-Shot Gravity Calibration

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Abstract

Predicting commuting flows without individual travel trajectories is a major challenge in urban planning. While deep learning models achieve high accuracy, they require localized origin-destination (OD) surveys for calibration, limiting deployment in data-scarce or privacy-restricted regions. This study shows that aggregate trip-length distributions (TLDs)—population-level travel distance histograms—are sufficient to recover city-specific distance-decay parameters without exposing individual trajectories. We evaluate an OD-free calibration setting using target-city TLDs (e.g., from Meta Movement Distribution Maps) together with open spatial covariates (primarily OpenStreetMap-derived features), and a pure zero-shot fallback using the same open covariates with a national prior. Across 50 US metropolitan areas, our aggregate-calibrated gravity model recovers decay parameters and achieves downstream performance statistically equivalent to the locally-calibrated traditional Tanner baseline within a 0.01 Common Part of Commuters (CPC) margin. Furthermore, a game-theoretic Shapley analysis shows that distance decay is the dominant predictive component within the investigated gravity framework, accounting for 85.8% of the total CPC gain. This provides an interpretable, privacy-preserving pathway for zero-shot urban flow prediction without local OD surveys.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Accurate prediction of origin–destination (OD) travel flows is fundamental to transportation planning, infrastructure investment, and accessibility analysis (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Reliable OD information enables evidence-based infrastructure prioritization, travel demand forecasting, and urban mobility policy analysis across cities of all sizes. Consequently, accurate OD estimation has long been a central problem in transportation science.

Yet acquiring the localized mobility data required to calibrate OD prediction models remains a major practical challenge. Household travel surveys require substantial financial and institutional resources, limiting their coverage to a small fraction of cities worldwide (Egu and Bonnel 2020; Hussain et al. 2021). Mobile-phone and GPS trajectory records, while richer in spatial and temporal detail, are frequently unavailable because of commercial licensing restrictions and increasingly strict privacy regulations. As a result, the majority of cities—particularly those in data-scarce regions—lack reliable local OD observations, and this challenge has motivated extensive research on estimating OD flows from incomplete or alternative mobility information.

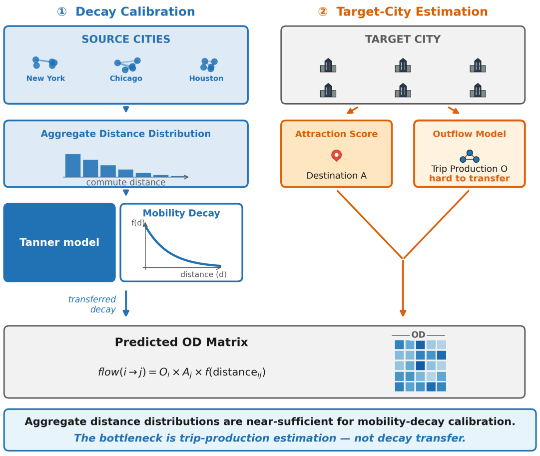

Fig. 1 Overview of the core findings: Aggregate trip-length distributions provide sufficient information for recovering city-specific distance decay $f(d)$, which acts as the dominant predictive component of gravity-based mobility prediction within the investigated framework. Prediction performance is bounded by the model's structural gravity assumptions.

A wide range of analytical and data-driven models have been developed to address this challenge. Classical spatial interaction models—including gravity models (Wilson 1971) and the radiation model (Simini et al. 2012)—are physically interpretable but require observed OD data to calibrate their distance-decay parameters. Learning-based approaches—including DeepGravity, MPGCN, and GODDAG (Simini et al. 2021; Hu et al. 2020; Rong et al. 2023a)—achieve high predictive accuracy by learning OD mappings from supervised flow matrices, imposing the same data dependency under a different computational guise. Transfer learning frameworks such as neuroGravity (Yang et al. 2026) and TransGM (Enaya et al. 2026) reduce but do not eliminate this dependency: they still require OD observations from a source city, a pretrained source-city model, or some degree of target-city structural data for adaptation. Despite their methodological diversity, these approaches all rely on some form of observed mobility information.

More precisely, existing OD prediction approaches ultimately depend on observed mobility data for model calibration, training, or transfer: gravity models require observed OD flows to calibrate distance-decay parameters; deep and graph learning methods require historical OD matrices for supervised learning; and transfer-learning approaches require mobility observations from source cities or auxiliary structural information for adaptation (Simini et al. 2021; Yang et al. 2026; Enaya et al. 2026). Although their modeling strategies differ substantially, they share the same underlying dependency on observed mobility information. This common limitation motivates the search for alternative mobility information that is both accessible and privacy-preserving.

Recent aggregate mobility products offer a promising new source of standardized, privacy-preserving mobility information. Meta's Movement Distribution Maps (MDM) (Meta 2026) provide aggregated population-level trip-length distributions (TLDs) rather than individual trajectories or OD matrices. These products are publicly available, globally consistent, and designed to protect user privacy, making them substantially more accessible in data-scarce and privacy-restricted regions (Pappalardo et al. 2023). However, whether aggregate trip-length distributions contain sufficient information to replace local OD observations in the calibration of gravity models remains an open question.

This question is grounded in a fundamental mechanism: distance decay—the well-documented tendency for spatial interaction to diminish with separation distance—is a dominant determinant of travel behavior (Tanner 1961; Verma and Ukkusuri 2025). Aggregate trip-length distributions are the observable statistical outcome of these distance-decay processes at the population level. This observation motivates the following scientific hypothesis: *aggregate trip-length distributions may preserve sufficient information to recover city-specific distance-decay functions without requiring local OD observations.* This hypothesis naturally raises a fundamental question regarding the recoverability of distance-decay information from aggregate mobility summaries.

It remains unclear, however, whether aggregate trip-length distributions contain sufficient information to recover expressive, city-specific distance-decay functions under realistic urban conditions. Prior work has demonstrated that simple aggregate statistics—such as median travel time—can calibrate single-parameter spatial interaction models (Merlin 2020), and that the form of distance-decay functions substantially affects spatial interaction outcomes (Lenormand et al. 2016). Whether this extends to the two-parameter Tanner deterrence function—requiring joint estimation of power-law and exponential components while correcting for zone geometry and attraction structure—remains uncharacterized. To address this unresolved question, this study investigates the following three research questions.

RQ1 Recoverability. Can aggregate trip-length distributions recover city-specific distance-decay parameters without access to any origin-destination observations?

RQ2 Information Content. If recoverable, how much predictive information is captured by the recovered distance-decay component relative to other gravity components?

RQ3 Practical Utility. Can the recovered distance-decay information support competitive survey-free OD prediction under strict zero-shot conditions (no target-city OD observations, no trajectory data, and no target-domain adaptation)?

This paper proposes **PIGF** (Projection–Inference Gravity Framework), a survey-free framework for gravity-model calibration using aggregate trip-length distributions. The proposed framework recovers city-specific distance-decay functions from aggregate TLDs and integrates them with transferable estimates of origin production and open-data-derived destination attraction for survey-free OD prediction. Our central claim is that *aggregate trip-length distributions preserve sufficient information to recover city-specific distance-decay functions, enabling accurate gravity-model calibration and survey-free OD prediction without local OD observations.* Our empirical investigation across 50 US metropolitan areas yields the following contributions:

1. We formulate gravity calibration as an exposure-corrected inverse problem and demonstrate that aggregate trip-length distributions can recover city-specific Tanner distance-decay parameters without any OD observations, achieving high parameter recovery fidelity across diverse urban morphologies.
2. We quantify the relative importance of gravity components through factorial ablation and game-theoretic Shapley analysis, finding that the distance-decay component accounts for 85.8% of the total predictive gain within the investigated framework—an empirical finding that motivates, rather than presupposes, its central role in our pipeline.
3. We develop a survey-free calibration pipeline (PIGF) that combines aggregate mobility summaries and open spatial data to achieve downstream performance statistically equivalent to locally-calibrated baselines, providing an interpretable and privacy-preserving pathway for zero-shot urban flow prediction.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the problem setting. Section 4 presents the methodological design. Section 5 reports experimental results. Section 6 discusses limitations and future directions, and Section 7 addresses threats to validity.

2 Related Work

2.1 Classical and Structured Mobility Models

Classical spatial interaction models, such as gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012), form the theoretical foundation of urban flow modeling. Gravity formulations relate traffic flows directly to zone masses and inversely to travel distance, while radiation models simulate destination choice based on the distribution of intervening opportunities. Although highly interpretable due to explicit physical constraints, gravity models require local origin-destination (OD) surveys to calibrate their distance-decay parameters (Martínez and Viegas 2013). The form and specification of distance-decay functions—ranging from exponential to power-law and composite deterrence functions such as the Tanner function—have long been debated in the transport geography literature (Lenormand et al. 2016; Feldman et al. 2012), with distance decay identified as the dominant determinant of spatial interaction strength across urban contexts (Verma and Ukkusuri 2025). Meanwhile, radiation models tend to perform poorly in urban regions characterized by highly directional commuting patterns or constrained transport networks.

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions.

Model Class	Decomposed Structure	Transferable Across Cities	Requires Target OD	Interpretable	Decay Explicitly Modeled
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD Prediction (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
PIGF (Ours)	Yes	Yes	No	High	Yes

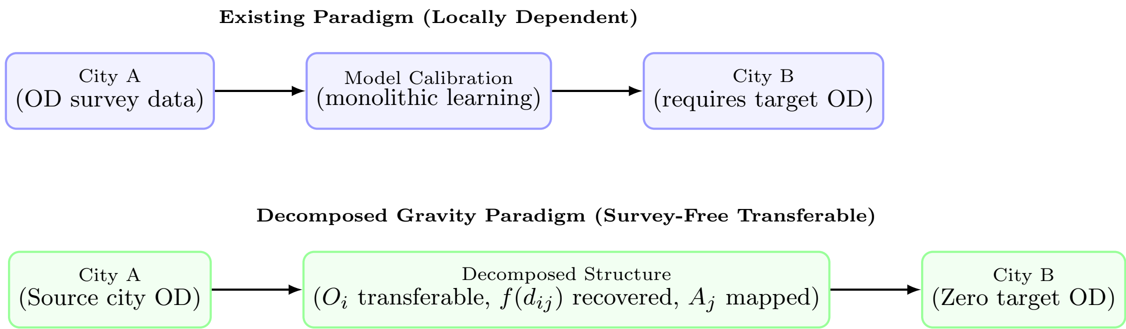

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling survey-free transfer without target-city OD surveys.

2.2 Deep and Graph Learning for OD Prediction

To overcome the limitations of classical gravity formulations, recent studies have increasingly adopted deep neural networks and graph neural networks (GNNs) to model complex spatial dependencies in OD prediction. Across this line of work, representative models include end-to-end neural formulations (e.g., DeepGravity), graph-convolutional OD predictors (e.g., MPGCN), graph-based transfer frameworks (e.g., GODDAG), universal geography neural networks, and OpenStreetMap-driven commuting predictors (Simini et al. 2021; Hu et al. 2020; Rong et al. 2023a; Guo et al. 2025; Atwal et al. 2025). The common pattern is that these methods learn OD mappings from supervised historical mobility data. As a result, they generally require historical OD matrices, trajectory-derived supervision, source-city OD data, or target-domain adaptation, which limits applicability under strict zero-shot deployment with no target-city OD observations.

2.3 Physics-Informed Neural Mobility Modeling

An additional emerging direction combines gravity-inspired structure with neural architectures. Representative studies include gravity-guided OD generation networks and hybrid gravity-gradient boosting models (Rong et al.

2023b; Liu et al. 2024). These approaches improve inductive bias by incorporating physical priors into learning pipelines. However, they still estimate flows from OD supervision rather than testing whether gravity parameters themselves can be recovered from aggregate mobility statistics alone.

2.4 Aggregate Mobility Data for Decay Calibration

To establish data-efficient modeling pipelines, researchers have explored aggregated mobility datasets that minimize privacy risks. These datasets can be compressed into trip-length distributions (TLDs), which represent one-dimensional marginal histograms of travel distances without disclosing individual zone-to-zone trajectories (Pappalardo et al. 2023). At the same time, cross-source differences in mobility data collection and preprocessing can significantly distort inferred displacement patterns and downstream conclusions (Gallotti et al. 2024).

Beyond privacy, aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—provide scalable summaries of displacement behavior (e.g., movement-distance histograms and distribution maps) that are increasingly available across regions and platforms. Existing literature primarily uses TLD-like statistics to characterize mobility patterns, analyze scaling laws, validate models, or provide auxiliary calibration constraints (Brockmann et al. 2006; Lenormand et al. 2016; Veneziano and Gonzalez 2011; Pappalardo et al. 2023). Recent studies also examine privacy-preserving synthetic mobility data pipelines and limitations of user-level differential privacy guarantees in aggregate location products (Ozaki et al. 2025; Houssiau et al. 2022). Notably, prior work has demonstrated that aggregate statistics such as median travel time can calibrate single-parameter spatial interaction models (Merlin 2020), suggesting that aggregate representations may contain sufficient information for gravity parameter identification. However, whether this extends to the two-parameter Tanner function remains under-characterized: even when aggregate shape statistics are informative, reliable inversion into city-specific distance-decay parameters requires controlling for geometry, attraction structure, and exposure effects.

2.5 Cross-City Transfer Learning

Cross-city transfer learning applies predictive knowledge from data-rich source cities to data-scarce target cities. Current transfer methodologies rely on region representation alignment (Wang et al. 2018), generative geographic pre-training (Mendieta et al. 2023), or domain adaptation (Wang et al. 2022). More recently, physics-informed deep learning frameworks such as neuroGravity (Yang et al. 2026) demonstrate that transferable mobility reconstruction from built-environment features is feasible without target-city OD observations. Likewise, TransGM (Enaya et al. 2026) proposes transferable gravity models for cross-city policy transfer by adapting a calibrated source-city gravity model to a target city. However, neuroGravity relies on complex learned neural representations, and TransGM still requires a previously calibrated source-city model and some degree of target-city structural data for adaptation. Neither approach investigates whether aggregate summary statistics alone—without any OD observations from source or target cities—are sufficient to identify city-specific distance-decay parameters.

2.6 Summary and Positioning

Table 1 compares how existing model classes address the core challenge of zero-shot cross-city transfer. Existing work broadly follows three paradigms: (i) learning OD flows directly from historical supervised data, (ii) transferring OD knowledge across cities through domain adaptation or learned representations (e.g., neuroGravity (Yang et al. 2026), TransGM (Enaya et al. 2026)), and (iii) integrating gravity-inspired priors into neural flow-learning models. In all three paradigms, OD observations or previously calibrated models remain central to training or adaptation.

In contrast, this study investigates a fundamentally different question: can aggregate trip-length distributions alone preserve sufficient information to recover city-specific distance-decay functions required for gravity-based OD generation—without any OD supervision? This formulates an *inverse mobility problem*: while recent work has learned deterrence functions directly from OD data via sparse regression

(Rubio-Herrero and Muñuzuri 2023), and aggregate statistics have been used to calibrate single-parameter gravity models (Merlin 2020), neither approach has investigated whether aggregate trip-length distributions alone can recover the two-parameter Tanner function without any OD supervision. To the best of our knowledge, existing literature has not demonstrated a practical workflow in which aggregate TLDs serve as the primary information source for recovering city-specific distance-decay parameters and then generating OD flows under strict zero-shot conditions. Accordingly, this work shifts the focus of zero-shot mobility modeling from learning OD flows to identifying the dominant transferable mechanism underlying those flows.

3 Problem Setting

We formalize the task of zero-shot cross-city travel flow prediction as follows. Let a target metropolitan area be partitioned into a set of N discrete spatial zones (e.g., census tracts) denoted by $\mathcal{V} = \{1, \ldots, N\}$. Our objective is to predict the origin-destination (OD) flow matrix $\mathbf{T} \in \mathbb{R}_+^{N \times N}$, where each element T_{ij} represents the volume of seasonal commuting trips from an origin zone $i \in \mathcal{V}$ to a destination zone $j \in \mathcal{V}$.

In a traditional local calibration setting, models are optimized using detailed individual-level trajectory records or comprehensive travel surveys that represent the target city's true flow matrix $\mathbf{T}$. However, in the zero-shot cross-city transfer paradigm, we assume that no local zone-to-zone flow observations are available for the target metropolitan area. The model must instead predict the target-city flows using only globally available built-environment features, gridded demographic data, and varying levels of aggregate target mobility signals.

To analyze the performance of urban mobility prediction under different data constraints and privacy regulations, we define three operational and diagnostic evaluation settings, which are summarized in Table 2. These settings represent distinct scenarios of data availability:

- **Pure Zero-Shot Fallback**: In this setting, the model relies entirely on open spatial features and a national distance-decay prior derived from source cities, assuming no target-city mobility data is accessible.
- **OD-Free Calibration**: In this primary proposed pipeline, the model has access to an aggregate, one-dimensional trip-length distribution (TLD) of the target city but cannot observe any individual origin-destination pairs.
- **Oracle Diagnostic Control**: Under this control setting, target-city origin outflows are set to their ground-truth values to isolate the errors of the distance-decay recovery from those of the outflow production predictions.

4 Methodological Design

The methodological design of this study is grounded in a separable formulation of the classical spatial gravity framework. Under this formulation, the commuter flow between an origin zone and a destination zone is modeled as a multiplicative product of distinct spatial components. This decomposition allows us to isolate the dominant components of spatial interaction from local, city-specific variables:

$$T_{ij} = O_i \cdot \frac{A_j f(d_{ij})}{\sum_k A_k f(d_{ik})} \tag{1}$$

where flows are decomposed into origin production (O_i), destination attraction (A_j), and distance decay ($f(d_{ij})$). We structure our investigation around three core hypotheses: (1) aggregate trip-length distributions (TLDs) contain sufficient information to recover city-specific distance-decay parameters without requiring individual origin-destination records; (2) the distance-decay component is the dominant predictive component within the investigated gravity framework; and (3) integrating recovered distance-decay parameters with machine learning models for production and open-data heuristics for attraction preserves strong survey-free target-city flow prediction utility.

4.1 Aggregate-Based Decay Calibration

We recover the distance-decay parameters $\theta = (\gamma, \beta)$ from the aggregate trip-length distribution (TLD) $p(d)$ — a marginal histogram of trip

Table 2 Operational and diagnostic evaluation settings.

Setting	Target City Mobility Data	Deployment / Analytical Purpose
Pure Zero-Shot Fallback	None (Open spatial features only)	Fallback scenario under extreme data scarcity; uses national decay prior.
OD-Free Calibration	Aggregate TLD (no individual OD flows)	Primary proposed pipeline; enables local calibration while protecting privacy.
Oracle Diagnostic Control	Target outflows O_i controlled	Diagnostic benchmark; isolates decay recovery from production prediction errors.

distances. Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \quad (2)$$

Because the trip-length distribution is conditioned by zone geometries and employment opportunity arrangements, we formalize the parameter recovery using an exposure-corrected likelihood model. Let y_b be the observed commuter count in distance bin $b \in \{1, \dots, K\}$ from the target-city TLD. Under the gravity framework, the predicted flow is:

$$\hat{T}_{ij}(\theta) = O_i \frac{A_j f_\theta(d_{ij})}{\sum_k A_k f_\theta(d_{ik})} \quad (3)$$

where O_i represents the target outflow, A_j is the attraction heuristic (Section 4.4), and f_θ is the distance-decay component. The predicted probability of a trip falling into bin b is:

$$P_\theta(b) = \frac{\sum_{i,j:d_{ij} \in b} \hat{T}_{ij}(\theta)}{\sum_{i,j} \hat{T}_{ij}(\theta)} \quad (4)$$

This predicted probability $P_\theta(b)$ explicitly corrects for the spatial opportunity distribution and zone geometries of the target city. Assuming a multinomial distribution of trips across the distance bins, we estimate the decay parameters $\theta = (\gamma, \beta)$ by maximizing the log-likelihood function defined over the observed fractions $\bar{y}_b = y_b / \sum_{b'} y_{b'}$ and predicted probabilities $P_\theta(b)$:

$$\mathcal{L}(\theta) = \sum_{b=1}^{K} \bar{y}_b \log P_\theta(b) \quad (5)$$

Minimizing $-\mathcal{L}(\theta)$ recovers the parameters from aggregate data without individual zone-to-zone records. In production, O_i in Eq. (3) is replaced by the GBDT-predicted $\hat{O}_i$; the model structure is shown in Fig. 3.

4.2 Distance-Decay Component

We model the distance-decay component using the Tanner multiplicative deterrence function (Tanner 1961; Feldman et al. 2012), which combines power-law and exponential terms and has been shown to outperform single-family alternatives across diverse urban settings (Verma and Ukkusuri 2025):

$$f(d_{ij}; \gamma, \beta, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \quad (6)$$

where $\beta > 0$ represents the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ serves as an adaptive geometric offset.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\frac{\text{Area}_i}{\pi}} \quad (7)$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. The parameters (γ, β) are estimated via MLE on the aggregate distance-bin distribution, with further optimization details provided in Appendix A. Because only binned trip-length histograms are used for calibration, this procedure is compatible with strict data-governance regulations (e.g., GDPR). For numerical implementation and computational efficiency, the theoretical offset δ is regularized by clamping the minimum distance to a small epsilon ($d_{\min} = 10^{-6}$), which yields mathematically equivalent results since $d_{ij} \gg d_{\min}$ for all cross-zone flows ($i \neq j$).

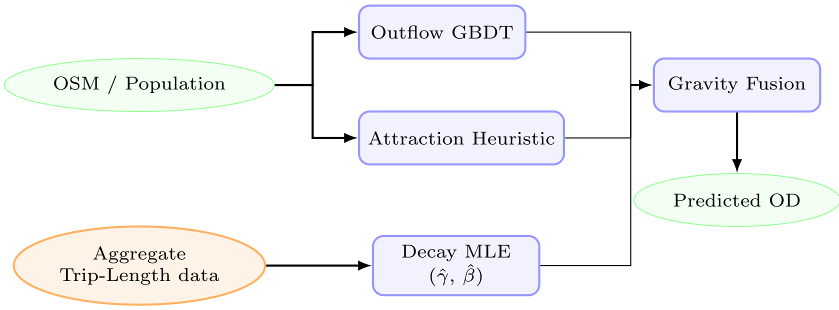

Fig. 3 Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—which provides aggregate summaries that significantly reduce privacy exposure compared with individual trajectories—feeds the Decay MLE block to recover city-specific parameters $(\hat{\gamma}, \hat{\beta})$ without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict survey-free flow matrices (T_{ij}).

4.3 Origin Production Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ represents the total commuter outflow originating from zone i. In zero-shot transfer settings where target outflows are unobserved, we predict O_i using a Gradient Boosted Decision Tree (GBDT) regressor. The GBDT model is trained on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i).

These six features were selected from an initial 37-candidate set using a cross-city SHAP stability analysis, showing no performance degradation compared to the full feature set. This parsimonious selection is grounded in the dimensional scaling of urban physics and Occam's razor, thereby preventing the model from overfitting to source-domain geometries. Hyperparameters and details of the SHAP-based feature selection are provided in Appendix C.

4.4 Destination Attraction Component (A_j)

The destination attraction A_j represents a zone's capacity to draw trips. To prevent the model from overfitting to source-domain geometries in zero-shot settings, we define a training-free, scale-invariant attraction heuristic:

$$\hat{A}_j = \frac{1}{2}\left[\mathrm{minmax}(P_j) + \mathrm{minmax}(\mathrm{POI}_j)\right] \tag{8}$$

where $\mathrm{minmax}(x_j) = (x_j - \min_k x_k)/(\max_k x_k - \min_k x_k + 10^{-9})$, P_j is the gridded population count, and POI_j is the OpenStreetMap POI count. This formulation normalizes absolute scale differences across cities while preserving local opportunity hierarchies. In Section 5.2, the ablation analysis (Table 6) validates that this training-free heuristic contributes significantly to the downstream prediction accuracy.

4.5 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4.6 Data Sources

We utilize three globally available open data sources to compile features for our models: built-environment spatial descriptors (OpenStreetMap POIs and road density), gridded demography (WorldPop), and aggregate movement telemetry maps (Meta Movement Distribution Maps (Meta 2026)). Ground-truth origin-destination (OD) flows between census tracts are obtained from the LODES commuting dataset covering 50 US metropolitan areas (25 source cities for GBDT production training and 25 held-out target cities for testing). The target OD matrices are kept completely blind to the model components and are used exclusively as ground truth for validation.

Algorithm 1: Aggregate Calibration and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate trip-length histogram $\{y_b\}_{b=1}^{K}$ (or national prior if unavailable)
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Offline Training (Source Cities)
1. **Production Model Training**: Train the outflow model $g(\mathbf{x})$ (GBDT) on pooled source-city spatial features $\mathbf{X}^{(c)}$ and ground-truth outflows $O_i^{(c)}$.

Phase 2: Online Calibration & Zero-Shot Transfer (Target City)
2. **Decay Calibration**: Recover target-city decay parameters $(\hat{\gamma}, \hat{\beta})$ via MLE on the target city's aggregate distance-bin histogram $\{y_b\}_{b=1}^{K}$. (If unavailable, fall back to the source-derived national prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$).
3. **Attraction Mapping**: Compute target city zone attractions using the training-free heuristic: $\hat{A}_j^{(t)} = \frac{1}{2}\left(\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})\right)$.
4. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   c. Fuse components and normalize outflows: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} \frac{\hat{A}_j^{(t)}(d_{ij}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}}}{\sum_k \hat{A}_k^{(t)}(d_{ik}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ik}}}$.

4.7 Calibration from Aggregate Trip-Length Distributions: Meta MDM Case Study

Under the proposed theoretical framework, any aggregate data source representing the probability distribution of human movement distances is sufficient to recover city-specific distance-decay parameters. In this paper, we use the term Trip-Length Distribution (TLD) to denote aggregate travel distance distributions, including products such as the HDX Movement Distribution dataset. To evaluate this formulation under real-world telemetry constraints, we utilize Meta's Movement Distribution Maps (MDM) as a representative case study of such platform-derived aggregate mobility indicators.

Meta MDM provides aggregate travel fractions (representing empirical TLDs) at the county level grouped into 3 physical distance bins: $[0, 10\text{ km}]$, $(10, 100\text{ km}]$, and > 100 km. To harmonize these county-level travel fractions into tract geometries for calibration:

1. County-level trip-length distributions are averaged using WorldPop-derived population shares as weights: $p_{\text{metro}}(b) = \sum_{c \in \mathcal{C}} w_c \cdot p_c(b)$, where w_c is the population share of intersecting county c.
2. Tract pairs (i, j) are mapped to these 3 bins based on their centroid distances d_{ij}.
3. The predicted probability of a flow falling into Meta bin b is computed by summing and normalizing predicted tract-level flows in that bin: $P_\theta(b) = \sum_{i,j:d_{ij} \in \text{Range}(b)} \hat{T}_{ij}(\theta) / \sum_{i,j} \hat{T}_{ij}(\theta)$. This normalization bridges the continuous gravity formulation with Meta's 3-bin telemetry.

4.8 Exploratory Urban Morphology Categorization

To investigate how physical geography modulates the role of distance decay in spatial interactions, the 25 target cities are classified into four morphological groups based on their urban structure and geography (González et al. 2008). This classification is summarized in Table 3.

5 Results

5.1 Can aggregate trip-length distributions reliably recover city-specific distance-decay parameters?

We begin by evaluating direct parameter recoverability from aggregate trip-length distributions (TLDs). The recovery metrics reveal different aspects of fit quality across the two decay exponents: the power-law exponent γ shows strong

Table 3 Morphological classification of target cities.

Morphology Group	Cities	Description
Dense / Transit ($n = 5$)	Boston, Long Beach, New York, Philadelphia, Washington DC	High-density urban cores; extensive transit networks; weaker distance decay influence.
Sprawling / Grid ($n = 10$)	Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa	Low-to-medium density; regular road grids; car-dependent mobility with uniform decay.
Polycentric ($n = 7$)	Albuquerque, Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose	Multiple employment centers; flows cluster around local nodes rather than a single center.
Coastal ($n = 3$)	Portland, Seattle, Virginia Beach	Coastline/topographic barriers constrain the road network, amplifying distance friction.

structural recovery across cities ($r = 0.909$), while the exponential rate β shows weaker cross-city rank correlation ($r = 0.427$) but remains close in absolute magnitude (MAE = 0.0026) (Table 4).

We then validate whether these recovered parameters preserve downstream predictive behavior. Under oracle outflow control, employing decay parameters recovered from aggregate TLDs yields OD prediction accuracy close to that obtained with parameters fitted on full zone-to-zone OD surveys (Table 5). Specifically, the average difference in the Common Part of Commuters (CPC) index between recovered and survey-calibrated parameters is a negligible 0.11%, with GBDT outflow prediction remaining the primary performance bottleneck in full zero-shot deployments (Table 7).

5.1.1 Validation via Trip-Length Distributions (Meta MDM Case Study)

To evaluate the practical feasibility of recovering distance decay from real-world TLDs, we apply our MLE framework to Meta's Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`) as an empirical case study across the 25 target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on LODES commuting data aggregated into 3 bins (`LODES Baseline (3-Bin)`) and 20 equal-width bins (`LODES Oracle (20-Bin)`).

As shown in Table 5, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities, compared to 0.7348 for the 3-bin survey baseline and 0.7403 for the 20-bin survey oracle.

Histogram compression is not the primary source of performance loss. When the original LODES commuting data is compressed from 20 bins to the same three physical distance bins used by Meta MDM (0–10 km, 10–100 km, > 100 km), CPC decreases only from 0.7403 to 0.7348 (−0.0055). This indicates that the coarse 3-bin representation preserves most of the information required for distance-decay recovery. In contrast, replacing the survey-derived histogram with real Meta MDM telemetry further reduces CPC from 0.7348 to 0.7080 (−0.0268). Therefore, most of the observed degradation is attributable to source mismatch and telemetry aggregation effects rather than to the limited number of bins. These results support the central claim that aggregate trip-length distributions remain highly informative for distance-decay calibration even under extremely compressed representations.

To evaluate performance under extreme data scarcity where target trip-length distributions (TLDs) are completely missing, we analyze the National Decay Fallback strategy. Under this strategy, the target city's distance-decay parameters are set to a national prior defined as the median parameter estimates fitted across the set of source cities. Under oracle outflows, this national prior model achieves a mean CPC of 0.7398, and under GBDT-predicted outflows, it yields a mean CPC of 0.6896 (as reported in Table 7).

Furthermore, we observe that the localized Meta MDM calibration is sensitive to specific local outlier dynamics. For instance, the metropolitan area of Atlanta serves as a prominent outlier in our evaluation, exhibiting a localized performance drop compared to the 3-bin LODES baseline.

Table 4 Decay parameter recovery performance metrics across the 25 target cities.

Parameter	Pearson r	MAE	Median Relative Error (%)	Cities within $\pm$10%	Practica tion
Power-law exponent (γ)	0.909	0.035	1.7%	100.0%	Strong st across cit
Exponential decay rate (β)	0.427	0.0026	0.0%	72.0%	Small despite w correlatio

Table 5 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies (Oracle Outflows).

Calibration Strategy	Data Source	Downstream CPC	CPC Gap vs. Oracle
Ground Truth Decay	Ground truth data	0.7411	Reference
20 equal width bins	TLD from GT data	0.7403	-0.0008
3 unequal width bins	TLD from GT data	0.7348	-0.0063
3 unequal width bins	Meta Movement Dist. Maps	0.7080	-0.0331

A detailed discussion of this outlier behavior is provided in Appendix A.2.

5.1.2 Decay Bin Sensitivity Analysis

To investigate how the granularity of distance binning affects model performance, we sweep the number of distance bins K across $\{3, 5, 10, 20\}$. The results show that predictive performance improves substantially from $K = 3$ to $K = 5$ and then exhibits diminishing returns, with mean CPC values of 0.7174 ($K = 3$), 0.7346 ($K = 5$), 0.7394 ($K = 10$), and 0.7403 ($K = 20$). Detailed comparisons and the corresponding sensitivity heatmap are provided in Appendix A.2 (Fig. 4).

5.2 To what extent does distance decay contribute to predictive performance within gravity-based mobility modeling?

To address this question on the extent of distance-decay contribution, we first evaluate the origin production component (O_i) in isolation to validate the predictive quality of our GBDT outflow models, and then implement a game-theoretic Shapley value analysis to quantify the relative contributions of each individual model component to the overall predictive accuracy.

5.2.1 Origin Production Transferability

To isolate the performance contribution of the origin production component O_i, we hold the destination attraction component A_j (computed via the OpenStreetMap heuristic) and the decay parameters (γ, β) (set to the national prior) fixed. Under this controlled setting, we compare three distinct outflow prediction strategies across the 25 target cities: a naive population baseline ($O_i \propto \text{Population}_i$), the proposed GBDT-predicted outflows, and the oracle outflows derived from ground-truth data.

The empirical results indicate that the GBDT outflow prediction model (which achieves a mean CPC of 0.6896) outperforms the naive population baseline (mean CPC of 0.6758). Furthermore, this GBDT configuration achieves 93.2% of the oracle upper bound (mean CPC of 0.7398) under zero-shot transfer conditions. Additional robustness experiments analyzing production transferability are discussed in Appendix B and illustrated in Fig. 5.

Feature Selection via SHAP Stability Analysis. To construct a parsimonious model, we evaluate the predictive performance of the GBDT regressor across nested subsets of the top features identified via SHAP stability analysis. This evaluation shows that the zero-shot prediction performance plateaus when using $K = 6$ macroscopic features. The detailed feature taxonomy, the SHAP stability selection criteria, and the corresponding feature count sensitivity analyses are provided in Appendix C (specifically in Fig. 6 and Fig. 7).

5.2.2 Factorial Ablation and Shapley Value Analysis

To evaluate the relative structural importance of each gravity model component in zero-shot transfer scenarios, we perform a factorial ablation study across the 25 target cities. Each component is disabled in turn and replaced by a uniform-weight fallback, which isolates its marginal contribution to the gravity model formulation.

Based on this ablation, we execute a game-theoretic analysis using Shapley values (Shapley 1953) computed across all $2^3 = 8$ possible component coalitions. Relative to a uniform-distribution baseline (assigning equal flow weight to all zone pairs; CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2258 (accounting for 85.8% of the total gain) to the distance-decay component $f(d)$, a gain of 0.0208 (7.9% of the total gain) to the attractiveness heuristic A_j, and a gain of 0.0167 (6.4% of the total gain) to the origin production component O_i. The baseline model for the factorial ablation presented in Table 6 is the complete gravity model utilizing the national decay prior (mean CPC of 0.6896), where removing the distance-decay component $f(d)$ causes the performance to drop to a CPC of 0.4489.

5.3 Can recovering distance decay from aggregate trip-length distributions enable competitive predictive performance without access to individual OD data?

To address the final research question, we evaluate whether our proposed aggregate-calibrated gravity model can serve as a deployable, survey-free alternative to traditional locally-calibrated frameworks. We compare our model against zero-shot neural architectures and locally calibrated structures across the 25 held-out target cities. The results of these comparative evaluations are summarized in Table 7.

Our most significant empirical result is that the proposed aggregate-calibrated model achieves statistical equivalence with the locally-calibrated Traditional Tanner baseline. Specifically, under oracle outflows, a Two One-Sided Test (TOST) of equivalence confirms that our model using Meta MDM decay (CPC 0.7337) is equivalent to the locally-fitted baseline (CPC 0.7382) within a tight margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$), despite utilizing only aggregate telemetry. A Wilcoxon signed-rank test shows that although the minor performance difference is statistically significant (raw $p = 0.00737$), the gap is practically negligible. Additionally, the proposed model statistically outperforms the parameter-free Radiation Model baseline (Wilcoxon test with Bonferroni adjustment, $p < 10^{-6}$).

Beyond this primary equivalence result, the multi-baseline comparison yields several supporting insights. First, we observe that predicting true origin outflows (O_i) does not significantly alter the predictive accuracy of the DeepGravity model (CPC 0.7526 vs. 0.7529), as its neural architecture jointly optimizes production and destination selection. In contrast, the proposed gravity model demonstrates a strong dependency on production accuracy, where the introduction of oracle outflows narrows the performance gap, nearly matching the performance of the locally-fitted baseline.

5.4 Exploratory Analysis of Urban Morphology Effects

Beyond evaluating overall predictive performance, we conduct an exploratory analysis to examine whether urban morphology may influence the effectiveness of aggregate-based distance-decay calibration. Given the limited number of target cities and the single-country scope of the dataset, these analyses are intended to generate hypotheses for future investigation rather than establish causal relationships.

Preliminary patterns suggest that the physics-constrained gravity formulation achieves its highest predictive performance in Coastal environments (mean CPC of 0.714). The relative performance gap between the proposed gravity model and the neural DeepGravity baseline narrows to 5.8% in Coastal cities, compared to a larger 10.3% gap in Dense / Transit environments. The

Table 6 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Contribution (%)
Distance Decay $f(d)$	-34.9%	0.2258	85.8%
Destination Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

Table 7 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Survey-Free Deployment (No target OD data)				
Proposed Model (Survey-Free, Meta MDM)	0.6860	0.0391	Decomposed	6
Proposed Model (Survey-Free, National Fallback)	0.6896	0.0384	Decomposed	6
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27
Group 2: Oracle Diagnostic (True outflows $O_i^{\mathbf{true}}$)				
Proposed Model (Oracle O_i, Meta MDM)	0.7337	0.0256	Decomposed	6
Proposed Model (Oracle O_i, National Prior)	0.7398	0.0248	Decomposed	6
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27
Group 3: Locally-Calibrated Baselines (Fitted on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

observations are consistent with physical geographical barriers restricting alternative routes and a greater explanatory role of distance decay in commuting flows. We note, however, that these morphology-specific findings should be interpreted cautiously due to several key statistical limitations: the overall sample size of target cities is small ($N = 25$), the morphology groups are highly unbalanced (e.g., only 3 Coastal cities and 5 Dense / Transit cities), and the resulting statistical power is limited.

To supplement this categorical classification, we analyze the continuous relationship between urban density and the performance gap ($\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$). The gap correlates positively with both median employment density ($r = 0.263$) and population density ($r = 0.252$). The results are consistent with the hypothesis that dense, transit-oriented development decouples urban trips from simple Euclidean distance, allowing deep learning models to capture complex non-linear feature interactions, whereas lower-density or physically constrained environments follow more predictable distance friction behavior.

In summary, these findings should be interpreted cautiously, but they suggest that our aggregate-calibrated model provides a statistically equivalent alternative to locally-fit structures in low-density or physically bounded cities, offering a practical deployment pathway when local surveys are unavailable.

6 Discussion

The three research questions follow a single logical chain: recoverability → importance → utility. Together, they address the gaps identified in Table 1. The following discussion synthesizes the established findings and outlines practical implications of our decomposed gravity framework.

Table 8 Survey-Free Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.0%
Polycentric	7	0.704	0.759	7.2%
Coastal	3	0.714	0.758	5.8%

6.1 Empirical Evidence and Deconvolution Mechanics

Our results across 50 US metropolitan areas show that aggregate trip-length histograms preserve sufficient signal to recover city-specific distance-decay parameters. The recovered parameters correlate strongly with the power-law exponent γ ($r = 0.909$) and more weakly with the exponential rate β ($r = 0.427$) fitted from complete OD flows. Under oracle outflows, our analysis indicates through a Two One-Sided Test (TOST) that the aggregate-calibrated gravity model is equivalent to the locally-calibrated Tanner model within a margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Additionally, Shapley analysis reveals that distance decay is the dominant predictive component within the investigated gravity framework, accounting for 85.8% of the total CPC gain.

The lower Pearson correlation for β should be interpreted in the context of scale. The cross-city numerical range of β is much narrower than that of γ, so small estimation deviations can substantially reduce correlation even when absolute errors remain negligible. Therefore, for β, absolute error metrics are more informative than rank correlation alone when assessing practical recovery quality.

While the importance of distance friction has long been recognized in mobility modeling, our results show that most predictive gain within the investigated gravity framework can be attributed to the distance-decay component. More importantly, distance-decay parameters can be recovered from aggregate trip-length distributions without requiring individual OD trajectories.

This recovery is explained by marginal projection under separability. Under separability, the OD matrix represents a joint distribution $P(X, Y)$ whose projection onto the distance domain D yields the TLD. Because the distance-decay component $f(d)$ is the only component depending explicitly on zone distances, this projection preserves its shape. Although the spatial layout of population and POIs distorts the marginal distribution, the exposure-corrected MLE (Equation 5) deconvolves this geometric bias to isolate distance-decay parameters.

Consequently, the TLD is not itself the dominant predictive component; rather, it is an observable projection from which the dominant predictive component can be identified.

This deconvolution explains the comparative behavior under different settings. Under oracle outflows, the ground-truth-derived national prior slightly outperforms the binned Meta MDM calibration (CPC 0.7398 vs. 0.7337). Under GBDT-predicted outflows, this gap narrows to 0.0036 CPC (CPC 0.6896 vs. 0.6860), as the continuous national prior acts as a regularizer, smoothing out localized flow fluctuations and reducing sensitivity to production prediction noise. Critically, this comparison highlights a key conceptual contribution: the objective of localized aggregate recovery is not to outperform a rich national prior in data-abundant regions (such as the United States), but rather to serve as a direct replacement where such a prior does not exist. Constructing a national prior requires extensive historical commuting surveys across multiple representative source cities, which is a luxury unavailable in developing nations or data-scarce territories. In these zero-data deployment scenarios, the global availability of platform-sourced telemetry (e.g., Meta MDM) offers an essential, training-free calibration signal that enables zero-shot flow prediction where it was previously impossible. Finally, we treat the role of urban structure and physical morphology not as a core validated finding, but as an exploratory observation that may help explain variation in model performance across cities.

6.2 Practical Deployment Implications

By demonstrating that high-fidelity decay calibration is achievable using only target-city aggregate histograms, this study enables a new paradigm for deploying mobility models. Calibrating per-city gravity models traditionally required invasive individual-level GPS telemetry or costly surveys. By combining globally available open-data features (OSM, WorldPop) with target-city aggregate TLDs, planning agencies can deploy structured gravity models in data-scarce or highly regulated environments. This pipeline bypasses the need for local travel surveys while preserving data confidentiality.

7 Threats to Validity

To ensure objective interpretation, we discuss the core limitations of our transfer learning workflow:

- **Target Mobility Dependency**: Under deployed GBDT outflows, the national prior fallback slightly outperforms localized calibration (CPC 0.6896 vs. 0.6860), indicating that localized calibration from binned telemetry does not necessarily yield downstream gains compared to a robust national prior. However, this concern is mitigated by the fact that a national prior cannot be constructed in data-sparse domains, leaving localized aggregate calibration as the only viable deployment pathway in such contexts.
- **Data Source Mismatches**: As demonstrated in Section 5.1, the performance gap is primarily driven by telemetry source mismatches and spatial aggregation rather than by the coarse 3-bin compression itself. The domain gap between survey-based commuting distributions and platform-derived mobility indicators remains a limitation for localized zero-shot calibration.
- **Temporal Stability Assumption**: Our calibration pipeline assumes a static trip-length distribution (TLD) for each city. Temporal stability of TLDs was not evaluated across weekday/weekend patterns, seasonal variation, epidemiological disruptions (e.g., COVID-19), or special-event disruptions.
- **Sample and Geographic Constraints**: The morphological classification contains small sample sizes (e.g., Coastal cities, $n = 3$). Furthermore, because our empirical evaluation is restricted to metropolitan areas in the United States, the observed commuting patterns, density profiles, and land-use arrangements may not directly generalize to European, Asian, or South American contexts.
- **Metric and Geometric Simplifications**: CPC is a global index blind to localized error structures, and using Euclidean distance ignores routing circuity and congestion travel times, which decouple travel from geometric distance in dense transit environments.

This study investigates the zero-shot cross-city transfer problem through a three-tier conceptual architecture, moving from ad-hoc calibration toward a structural analysis of urban spatial interaction. Our findings across 50 US metropolitan areas indicate that distance decay is the dominant predictive component in the evaluated gravity framework. We also show that the aggregate trip-length distribution (TLD) provides enough information to recover this component for zero-shot gravity model calibration.

7.1 Summary of Empirical Findings

We demonstrate that target-city aggregate TLDs contain sufficient signal to recover distance-decay parameters, with downstream prediction performance statistically equivalent to models calibrated directly on target-city individual-level OD flows. A game-theoretic Shapley analysis shows that distance decay accounts for 85.8% of the total CPC gain, making it the dominant predictive component in the evaluated framework. Although localized calibration from real-world telemetry does not outperform a robust national decay fallback, it does not require historical survey data from source cities, thereby enabling zero-shot deployment in data-scarce regions where such national priors cannot be constructed.

7.2 Theoretical and Practical Implications

These findings redefine how spatial interaction models can be calibrated. Historically, identifying city-specific travel friction required highly sensitive individual-level surveys or proprietary cellular trajectories. By demonstrating that distance-decay parameters can be robustly recovered using a one-dimensional aggregate representation (the TLD), we provide a deployable, survey-free forecasting pipeline. This architecture enables the calibration of structured, interpretable gravity models using only globally available open spatial data (OSM, WorldPop) and privacy-compliant aggregate indicators, facilitating deployment under privacy constraints and reducing survey costs.

7.3 Limitations and Future Directions

Despite these contributions, several challenges remain. First, the separability approximation assumes independent spatial components, which may exhibit non-linear coupling in reality. Second, the empirical evidence is restricted to US metropolitan areas characterized by high private vehicle usage; transferability to highly dense, multi-modal transit networks or informal settlements requires further cross-regional verification. Finally, using Euclidean distance ignores network routing circuity and congestion travel times. Incorporating these network and temporal dynamics into the aggregate calibration process represents an essential direction for future work.

A particularly promising direction concerns the role of urban morphology. The exploratory analyses suggest that physical urban structure may influence the effectiveness of aggregate-based decay calibration. However, the current evaluation contains only 25 target cities from a single country, limiting statistical power and geographic diversity. Future work should evaluate substantially larger cross-country datasets spanning diverse urban forms, transportation systems, and development contexts to determine whether morphology-specific calibration strategies are warranted.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

To solve the maximum likelihood estimation (MLE) problem for target-city decay parameters $\theta = (\gamma, \beta)$ without disaggregate flows, we adopt a robust and physically grounded initialization and optimization workflow based on closed-form moment estimates.

Good parameter initialization is critical for optimization convergence. For the exponential decay rate β, we utilize a closed-form moment estimate derived under the assumption of pure exponential deterrence and a uniform distribution of spatial opportunities. In this theoretical limit, the expected travel distance satisfies $\mathbb{E}[d] = 1/\beta$. We estimate the empirical mean trip distance $\bar{d}_{\text{obs}}$ from the observed distance-bin histogram:

$$\bar{d}_{\text{obs}} = \sum_{k=1}^{K} b_k \cdot m_k \tag{9}$$

where b_k is the observed proportion of trips in bin k, and m_k is the midpoint of distance bin k (for the open-ended final bin, we use 1.5 times the lower boundary as a surrogate midpoint). The initial value $\beta^{(0)}$ is set as:

$$\beta^{(0)} = \frac{1}{\bar{d}_{\text{obs}}} \tag{10}$$

For the power-law exponent γ, the initial value is set statically to $\gamma^{(0)} = 1.0$, which corresponds to the center of the empirically observed range $[0.5, 2.0]$ in classical spatial interaction modeling literature.

The positivity constraints $\beta > 0$ and $\gamma \geq 0$ are strictly enforced during optimization via a parameter transformation. We define log-transformed parameters:

$$\tilde{\beta} = \log \beta, \quad \tilde{\gamma} = \log \gamma \tag{11}$$

and optimize the objective in the unconstrained log-transformed space $\theta_{\log} = (\tilde{\gamma}, \tilde{\beta}) \in \mathbb{R}^2$.

To guard against convergence to local optima due to the potential non-convexity of the exposure-corrected likelihood surface (especially under highly binned or noisy data), we run the L-BFGS-B optimization algorithm from 10 random restarts. The first run is initialized at the theoretical values $\theta_{\log}^{(0)} = (\log \gamma^{(0)}, \log \beta^{(0)})$. The remaining 9 runs are initialized by perturbing this central starting point with Gaussian noise:

$$\begin{aligned} \tilde{\gamma}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \gamma^{(0)}, 0.25), \\ \tilde{\beta}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \beta^{(0)}, 0.25) \end{aligned} \tag{12}$$

The optimization is executed for each restart run, and the parameter set yielding the highest likelihood value $\mathcal{L}(\theta)$ (Equation 5) is recovered via exponential back-transformation: $\hat{\gamma} = \exp(\tilde{\gamma}^*)$ and $\hat{\beta} = \exp(\tilde{\beta}^*)$.

A.2 Robustness Analysis

To evaluate the reliability of the exposure-corrected maximum likelihood estimation (MLE) workflow under non-ideal data collection conditions, we conduct a series of robustness analyses across four scenarios:

- **Bin Sensitivity Analysis**: Sweeping the number of distance bins $K \in \{3, 5, 10, 20\}$ reveals that downstream prediction performance stabilizes quickly. The mean CPC across target cities increases from 0.7174 under a coarse $K = 3$ configuration to 0.7346 for $K = 5$, 0.7394 for $K = 10$, and 0.7403 for $K = 20$ (compared to a baseline of 0.7411 computed on full, unbinned ground-truth OD matrices). This suggests that five or more distance bins provide sufficient resolution to capture city-specific decay profiles.
- **Atlanta Case Study Outlier**: Atlanta is identified as a localized outlier, where its Meta MDM 3-bin CPC under oracle outflows is 0.5761 compared to 0.7334 under the LODES 3-bin baseline. This drop is driven by the spatial interaction between the sprawling polycentric layout of the city and the coarse telemetry binning of the Meta Movement Distribution Maps (MDM) data. Excluding Atlanta from the target set increases the mean target CPC for Meta MDM from 0.7080 to 0.7135, recovering 96.4% of the corresponding LODES baseline.

- **Missing Attractiveness Data (A_j)**: We simulate spatial feature scarcity by randomly removing 10%, 20%, and 50% of target-city destination attraction values A_j. The parameter recovery remains stable, yielding a mean absolute error (MAE) of $\gamma_{\text{MAE}} = 0.0395$ under 20% missing data, compared to the baseline MAE of 0.0394.
- **CBD Collapse (Extreme A_j Distortion)**: To test resilience against extreme structural distortions, we replace the attraction values A_j of the top 10% most attractive zones with the city-wide mean attraction. Under this CBD collapse scenario, the estimation remains robust, yielding parameter recovery errors of $\gamma_{\text{MAE}} = 0.0441$ under direct replacement and $\gamma_{\text{MAE}} = 0.0486$ when the affected zone attractions are scaled down by 80%.

A.3 Preprocessing and Normalization

Data preprocessing details are summarized in Table 9.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of the zero-shot origin production transfer along two operational dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot outflow predictability by retraining the GBDT model on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Evaluating on target cities demonstrates that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Under substantial injected noise (40%), the target CPC degrades from 0.6896 to 0.6626. While this -3.9% drop is larger than the isolated production component contribution (3.6% Shapley value), it demonstrates that even highly corrupted outflow inputs do not collapse the spatial interaction matrix, because the production-constraint normalization restricts the error propagation.

Appendix C. Feature Selection Sensitivity

C.1 Feature Taxonomy and Candidate Set (37 dimensions)

We initially construct 37 candidate features per zone. This candidate space comprises 5 base scale and density variables (zone area, road density, residential population, total POI counts, and POI density) and 32 POI category count and density features (16 category counts and 16 category densities). The feature taxonomy is shown in Fig. 6.

C.2 SHAP-Based Feature Stability Selection

To identify features that generalize across cities, we apply SHAP importance analysis (Lundberg and Lee 2017) to the GBDT production model. Features are ranked by a cross-city stability score $S_p = \mu_p \,/\, (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This analysis identifies 25 stable features from the 37-candidate space (32.4% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 4.3) are among the primary and most stable predictors, with no accuracy loss relative to the full candidate set. The GBDT hyperparameter configurations are detailed in Table 10.

C.3 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 25, 37\}$) shows that predictive performance plateaus at $K = 6$ features (CPC: 0.7080), with additional features yielding marginal utility (e.g., CPC of 0.7120 at $K = 15$). This indicates that most transferable spatial information is

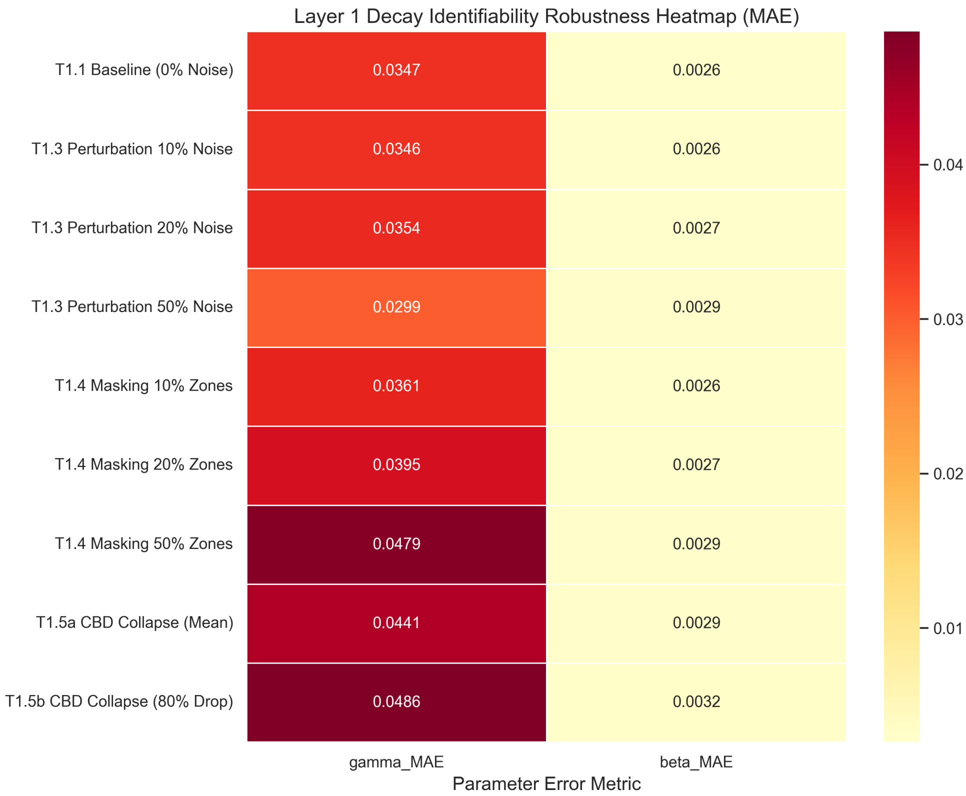

Fig. 4 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

Table 9 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	**Treatment**	**Rationale**
Missing Data	Median imputation	Rare ($<$ 2%); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

captured by a compact set of built-environment indicators.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 11.

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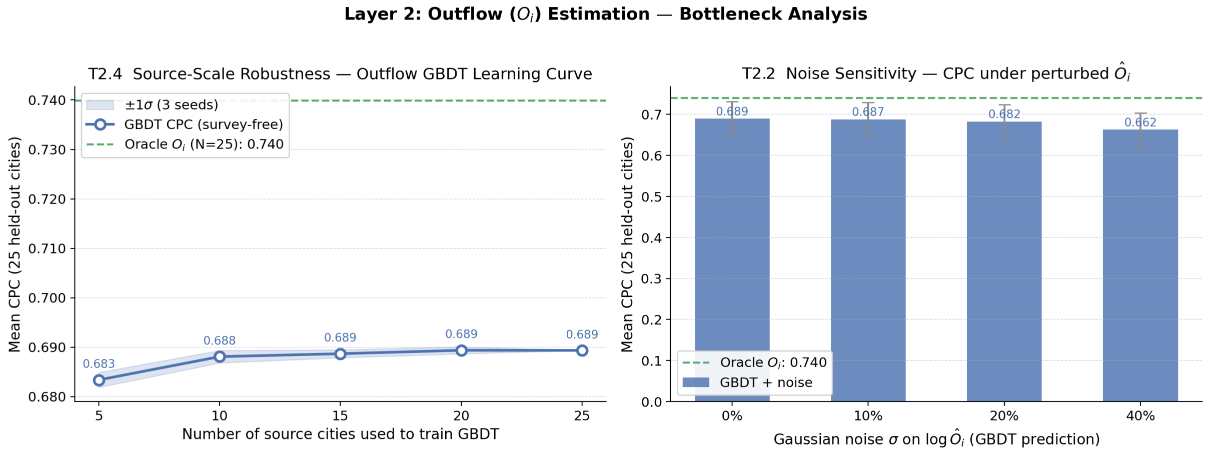

Fig. 5 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

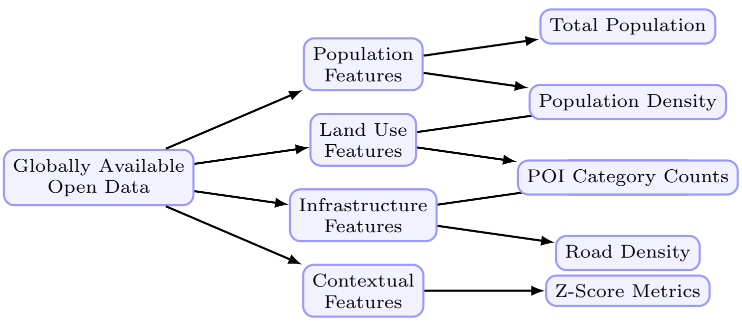

Fig. 6 Taxonomy of the globally available open data features used in the outflow model.

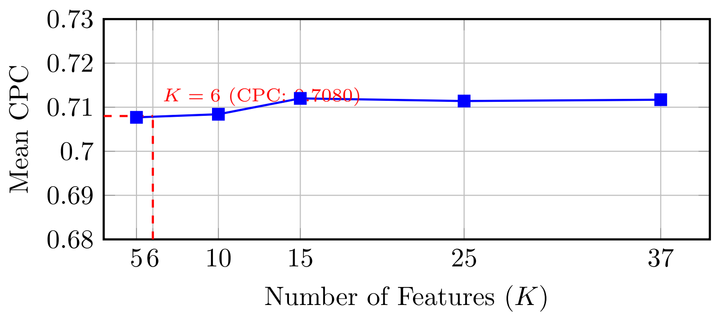

Fig. 7 SHAP feature count sweep over the 37-candidate feature set. While utilizing more than 6 features yields marginal accuracy gains, a parsimonious selection of 6 features is preferred to maintain generalization across diverse target-city environments.

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Table 10 GBDT Model Hyperparameter Configurations

Hyperparameter	Outflow Model (O_i)	Attraction Model (A_j, ablation only)
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	2	2
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Table 11 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations
PIGF	Projection–Inference Gravity Framework		

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